

Term structure determinants of time-varying risk of 1-year bond returns

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Abstract

Term structure drivers of 1-year bond premia and conditional bond return risk are distinct. Consequently, the Cochrane–Piazzesi factor captures aggregate price of risk and not the amount of risk in 1-year bond returns. One linear combination of forward rates captures most of the variation in bond return risk across maturities. Interest rate level captures substantial amount of variation in the conditional return risk, a finding consistent with rising inflation uncertainty with level of inflation and interest rates. The 4-5 yield spread, an important positive predictor of bond return premia, has an opposing but limited impact on the conditional volatility.

KEYWORDS

bond return volatility, interest rate volatility, risk management, yield curve

JEL CLASSIFICATIONS

E43, E44, G12, G17, G19

1 | INTRODUCTION

Literature provides strong evidence in favor of time variation in expected excess returns of bonds. Fama and Bliss (1987), Campbell and Shiller (1991), and Cochrane and Piazzesi (2005) show that linear combinations of bond yields or forward rates predict excess bond returns. Cochrane and Piazzesi (2005) provide strong evidence in favor of common component in the time variation of expected excess returns across maturities. They find that a single tent-shaped linear combination of forward rates captures most of the variation in the expected 1-year excess returns of bonds of 2- to 5-year maturities. The tent-shaped factor is able to predict the excess returns with more than 30% R^2 . Also, Fama and French (1989) find a common component in the time variation of expected excess returns on bonds and stocks. These papers suggest that the time variation in the expected excess returns of bonds is explained, at least partially, by a time-varying aggregate price of risk. In addition, Wachter (2006) shows that external habit preferences can generate a positive forecasting relation between the yield spread and bond excess returns.

Cochrane and Piazzesi (2005) suggest their tent-shaped Cochrane–Piazzesi factor, the driver of conditional mean of 1-year excess bond returns, to be a strong candidate for aggregate price of risk. However, note that equilibrium

asset pricing models obtain variation in the expected excess returns as a result of variation in the aggregate/market price of risk or variation in the amount of risk associated with investment in the asset or both. This precludes a conclusive interpretation of the driver of the expected excess returns (the driver of conditional mean) as the market price of risk. The Cochrane–Piazzesi factor, the common component in conditional mean, could in fact be proxying variation in the amount of risk or, in other words, the conditional volatility. In fact, Boyd, Hu, and Jagannathan (2005) and Andersen, Bollerslev, Diebold, and Vega (2007) show that stock returns, interest rates, and exchange rates have different responses to macroeconomic news over the business cycle, thus suggesting the existence of a business cycle component to the variation in the second moments of returns.

In this paper, I extract the term structure drivers of both the conditional volatility and conditional mean of 1-year excess bond returns and compare the two. I follow Cochrane and Piazzesi (2005) and conduct these tests for bonds of 2- to 5-year maturities. I show that information in the term structure explains a substantial amount of variation in the conditional volatilities of excess returns. I also show that a single linear combination of forward rates captures most of the time variation in the conditional volatilities of excess returns across maturities. This single factor is, however, different from the tent-shaped Cochrane–Piazzesi factor that drives the conditional mean. I test the extent to which the variation in the expected excess returns is due to the variation in the conditional volatility. I test the hypothesis that the same linear combination of forward rates drives both the conditional mean and the conditional volatility. The test reveals that such a restriction severely impedes the ability to forecast the conditional volatility (and produces a model that is a significant misfit to the data). I therefore conclude that the term structure drivers of the conditional mean and the conditional volatility are not the same. This shows that the tent-shaped Cochrane–Piazzesi factor that drives the conditional mean captures the market price of risk and not the conditional volatility, and as a result, this factor can be used for pricing the cross-section of asset returns.

I use forward rates as the primitives representing the term structure information. The specifications in terms of log conditional volatilities help avoid restrictions on the coefficients of the forward rates. The conditional volatility specification in terms of the short rate and the spreads of the forward rates (relative to the short rate) reveals that the overall level of interest rates captures most of the variation in conditional volatilities. The conditional volatility specification without the short rate is strongly rejected. In addition, spreads do also contribute a nontrivial but smaller amount to the variation of the conditional volatility. These findings are consistent with dominance of inflation uncertainty for determining the volatility of excess bond returns.

I distill the information in the term structure for forecasting the conditional volatility into two components: (1) the short rate; and (2) a factor formed from the spreads. The spread factor is a linear combination of the spreads of 5- and 4-year forward rates relative to the short rate. The loading is excessively tilted toward a positive contribution by the 5-year forward rate spread. Cochrane and Piazzesi (2005) note that the 4-5 yield spread is important for forecasting the 1-year excess returns of all maturities. The same 4-5 yield spread has an opposing effect on the conditional volatility. However, the short rate drives both the expected excess returns and the conditional volatility in the same direction, though the short rate dominates the conditional volatility variation. The opposing effects of spreads and the dominance of short rate for the conditional volatility variation leave the drivers of the conditional mean and the conditional volatility out of sync.

Glosten, Jagannathan, and Runkle (1993) suggest that the short rate reflects aggregate economic uncertainty in addition to expectations of future inflation and the real interest rate. Kim and Nelson (1999) and McConnell and Perez-Quiros (2000) show that the post-1981 period is characterized by significantly lower real and nominal macroeconomic volatility. This shift in macroeconomic volatility does not affect the ability of the short rate to forecast the conditional volatility.

The study closest to this paper is Viceira (2012). Viceira (2012) focuses on only the realized second moments of returns at 3- to 12-month horizons on a bond with a fixed 5 years to maturity and the time variation in business cycle proxies. Note that these realized volatilities do not correspond to volatilities of 1-year bond returns, the main subject of this paper. I focus on the 1-year excess returns of bonds of 2- to 5-year maturities for which Cochrane and Piazzesi (2005) find compelling evidence in favor of a common component in the expected excess returns. Viceira (2012) uses

daily returns on constant maturity bond that does not add up to 1-year holding period return on a bond. Consequently, results in Viceira (2012) do not address the question about whether it is the aggregate price of risk or the amount of risk associated with 1-year investment in bonds is reflected in the Cochrane–Piazzesi factor. Further, I use a maximum likelihood approach to arrive at estimates for the conditional mean and conditional volatility using information in the entire term structure.

The rest of the paper is organized as follows. Section 2 describes the data, and Section 3 describes the notation. Section 4 shows that the term structure captures a substantial amount of variation in the conditional volatility of excess bond returns. The tests in this section also show that the drivers of conditional mean and conditional volatility are different. The tests in Section 5 show that the level of interest rates captures most of the variation in the conditional volatility and that the spreads are responsible for only a small part of the variation. Section 6 presents evidence indicating that the spreads capture a small but nontrivial amount of the variation in the conditional volatility. Section 7 shows that the differences in macroeconomic volatility pre- and post-1981 do not alter the conclusions. Section 8 shows that the results in this paper are robust to including additional lagged values of conditional volatility as explanatory variables. Section 9 concludes the paper.

2 | DATA

I use monthly bond price data for bonds of 1- to 5-year maturity from the CRSP Fama-Bliss Discount Bond file. This data set of zero-coupon bond prices is created by bootstrapping price data of Treasury bills, notes, and bonds. This procedure creates hypothetical prices of zero-coupon bonds as a substitute for actual traded prices. Additional details are available in Fama and Bliss (1987). In this paper, I follow Cochrane and Piazzesi (2005) and use only the data from January 1964 onward. The sample spans the period from January 1964 to December 2018.

3 | NOTATION

I use following notation for the bond prices, returns, and forward rates:

$$p_t^{(n)} = \text{Log price of the } n\text{-year discount bond at time } t. \quad (1)$$

The log yield of the n -year discount bond at time t is:

$$y_t^{(n)} \equiv -\frac{1}{n} p_t^{(n)}. \quad (2)$$

The unit time interval throughout the paper is 1 year. Less than a year's time intervals are represented in fractions of 12.

The log forward rate at time t for borrowing between time $t + n - 1$ and $t + n$ is:

$$f_t^{n-1 \rightarrow n} = f_t^{(n)} \equiv p_t^{(n-1)} - p_t^{(n)}. \quad (3)$$

The log holding period return for buying an n -year bond at time t and selling it as an $(n - 1)$ -year bond at time $t + 1$ is:

$$hpr_{t+1}^{(n)} \equiv \frac{p_{t+1}^{(n-1)} - p_t^{(n)}}{1}. \quad (4)$$

The 1-year excess log return is:

$$rx_{t+1}^{(n)} \equiv hpr_{t+1}^{(n)} - y_t^{(1)}. \quad (5)$$

I denote the conditional variance of the 1-year excess return of an n -year maturity bond with respect to the time t information set as $h_{t+1}^{(n)}$ —that is, $\text{var}_t(r_{t+1}^{(n)}) \equiv h_{t+1}^{(n)}$. Note that the time index on the conditional volatility is $t + 1$, though the conditional volatility is measurable with respect to time t information. I use variations of the following two shorthand notations to succinctly denote certain sums throughout the paper:

$$\gamma_{1,\dots,5}^{(n)} \cdot f_t^{(1,\dots,5)} = \sum_{j=1}^5 \gamma_j^{(n)} \times f_t^{(j)}, \quad (6)$$

$$\beta_{2,\dots,5}^{(n)} \cdot (f_t^{(2,\dots,5)} - f_t^{(1)}) = \sum_{j=2}^5 \beta_j^{(n)} \times (f_t^{(j)} - f_t^{(1)}). \quad (7)$$

$\gamma_t^{(n)}$ and $\beta_t^{(n)}$ are the coefficients in different specifications. Note also that the conditional volatility of 1-year excess return on bonds maps one to one to the yearly volatility of yields. Therefore, the conclusions in the following sections also apply to yearly volatility of yields:

$$\sigma_t(r_{t+1}^{(n)}) = (n-1) \times \sigma_t(v_{t+1}^{(n-1)}), \quad (8)$$

$$\Rightarrow \sigma_t(v_{t+1}^{(n-1)}) = \frac{1}{n-1} \times \sigma_t(r_{t+1}^{(n)}) = \frac{1}{n-1} \times \sqrt{h_{t+1}^{(n)}}. \quad (9)$$

4 | INFORMATION IN THE TERM STRUCTURE

In this section, I test the hypothesis that the term structure captures a significant amount of the variation in the conditional volatility. The conditional volatilities of bonds of 2- to 5-year maturities are estimated using the maximum likelihood method. I find that a single linear combination of the forward rates captures most of the variation in the conditional volatilities due to variation in the forward rates. Further, I find that the term structure drivers of the conditional mean and the conditional volatility are not the same.

$$(S): \quad r_{t+1}^{(n)} = \kappa^{(n)} + \gamma_{1,\dots,5}^{(n)} \cdot f_t^{(1,\dots,5)} + \epsilon_{t+1}^{(n)} \quad \forall n = 2, \dots, 5, \quad (10)$$

$$\epsilon_{t+1}^{(n)} = \sqrt{h_{t+1}^{(n)}} \cdot u_{t+1}^{(n)} \quad \text{with } u_{t+1}^{(n)} \sim N(0, 1), \quad (11)$$

$$(S-0): \quad h_{t+1}^{(n)} = \exp \left(\alpha^{(n)} + \delta_1^{(n)} \ln \left(\left(\epsilon_t^{(n)} \right)^2 + \lambda \right) \right), \quad (12)$$

$$(S-1): \quad h_{t+1}^{(n)} = \exp \left(\alpha^{(n)} + \delta_1^{(n)} \ln \left(\left(\epsilon_t^{(n)} \right)^2 + \lambda \right) + \beta_{1,\dots,5}^{(n)} \cdot f_t^{(1,\dots,5)} \right). \quad (13)$$

Specification (S) relates the term structure information in the form of a linear combination of the forward rates to 1-year excess returns of bonds of 2- to 5-year maturities. Specification (S-1) relates the forward rates to the log of the conditional volatility. The log transformation avoids restrictions on the coefficients of the forward rates. I use a maximum likelihood approach to estimate the parameters in the conditional mean and the conditional volatility specifications. In all of the following specifications, λ is not an estimated parameter but a small constant to ensure the gradient of the likelihood function exists.¹ I use overlapping monthly data of 1-year excess returns of bonds of

¹The value of λ is set to 10^{-8} in all the specifications. The unexpected excess returns in (S-1) are raw returns without the conversion to percentage points. Setting λ to lower values up to 10^{-12} results in a higher number of iterations until convergence. The parameter estimates and the standard errors are not much different. $-\frac{2\epsilon_t^{(n)}\delta}{(\epsilon_t^{(n)})^2 + \lambda} [1, f_{t-1}^{(1,\dots,5)}]'$ is a part of the gradient of the likelihood function. For $\lambda = 0$ and zero or too small unexpected returns (relative to machine precision), the multiple is not defined.

TABLE 1 The χ^2 statistic for the likelihood ratio test of specification (S-0) with just the lagged residuals against the alternate specification (S-1), which also includes the forward rates

$H_0: S-0, H_1: S-1$			
	χ^2	p-Value	1% Crit- χ^2
$n = 2$	50.19	.00	15.09
$n = 3$	52.88	.00	15.09
$n = 4$	53.38	.00	15.09
$n = 5$	62.12	.00	15.09

TABLE 2 The coefficient estimates and the t -ratios in parentheses for the parameters in the conditional mean specification (S). The specification comprises of the pair (S), (S-1), the conditional mean and conditional volatility specifications, respectively

$rX_{t+1}^{(n)} = \kappa^{(n)} + \gamma_{1,\dots,5}^{(n)} \cdot f_t^{(1,\dots,5)} + \epsilon_{t+1}^{(n)}$						
	$\kappa^{(n)}$	$\gamma_1^{(n)}$	$\gamma_2^{(n)}$	$\gamma_3^{(n)}$	$\gamma_4^{(n)}$	$\gamma_5^{(n)}$
$n = 2$	−.02 (−5.0)	−.83 (−6.1)	.15 (.6)	1.19 (5.0)	.46 (2.4)	−.76 (−5.0)
$n = 3$	−.03 (−4.9)	−1.72 (−7.7)	−.02 (−.0)	3.09 (7.7)	.59 (1.8)	−1.64 (−6.8)
$n = 4$	−.04 (−5.3)	−2.20 (−6.9)	−.15 (−.2)	3.59 (6.1)	1.54 (3.2)	−2.35 (−6.5)
$n = 5$	−.05 (−5.5)	−2.88 (−7.5)	.26 (.4)	3.97 (5.5)	1.53 (2.6)	−2.35 (−5.4)

2- to 5-year maturities. The details of the estimation procedure and the method for correcting the errors are in the Appendix in Supporting Information.

One simple way to gauge the importance of forward rates for the conditional volatility estimation is to compare specifications (S-0) and (S-1). (S-0) contains just the lagged residual, whereas (S-1) also includes all the forward rates. The likelihood ratio test statistics in Table 1 imply that the term structure of interest rates makes a significant contribution toward predicting the conditional volatility of bond excess returns. The log-likelihood values for specification (S-1) far exceed the log-likelihood values for specification (S-0), which is devoid of forward rates.

The likelihood ratio test of specification (S-1) with and without the log of lagged conditional variance implies that the zero coefficient restriction on the log of the lagged conditional variance cannot be rejected. I discuss the conventional autoregressive conditional heteroskedasticity/generalized autoregressive conditional heteroskedasticity (ARCH/GARCH) specifications and the specification with the lagged conditional volatility in Section 8. Including the lagged conditional variance does not significantly improve the model's success at capturing the conditional volatility. For the rest of the paper, I do not include the lagged conditional variances in the specifications.

Tables 2 and 3 present the coefficient estimates and the standard errors for the parameters in specification (S-1). None of the coefficients of the forward rates in the conditional volatility specification are significant. Small t -ratios and large coefficients are a sign of possible near-multicollinearity. I take up the task of reducing the number of explanatory variables in the next few sections. However, the joint significance test rejects the hypothesis that the coefficients of the forward rates in the conditional volatility specification are zero. The χ^2 statistics in Table 4 reject the null even at the 1% significance level. Thus, the information in the term structure captures a nontrivial component of the variation

TABLE 3 The coefficient estimates and the t -ratios in parentheses for the parameters in the conditional volatility specification (S-1). The specification comprises of the pair (S), (S-1), the conditional mean and conditional volatility specifications, respectively

	$\ln(h_{t+1}^{(n)}) = \alpha^{(n)} + \delta_1^{(n)} \ln\left(\left(\epsilon_t^{(n)}\right)^2 + \lambda\right) + \beta_{1,\dots,5}^{(n)} \cdot f_t^{(1,\dots,5)}$						
	$\alpha^{(n)}$	$\delta_1^{(n)}$	$\beta_1^{(n)}$	$\beta_2^{(n)}$	$\beta_3^{(n)}$	$\beta_4^{(n)}$	$\beta_5^{(n)}$
$n = 2$	-10.03 (-17.8)	-.04 (-1.1)	5.76 (.4)	11.71 (.5)	4.75 (.2)	-13.36 (-.7)	9.13 (.7)
$n = 3$	-9.17 (-20.2)	-.06 (-2.0)	-9.38 (-.7)	23.99 (.9)	5.15 (.2)	-12.80 (-.7)	12.32 (.9)
$n = 4$	-8.19 (-16.7)	-.02 (-.5)	-15.54 (-1.2)	28.77 (1.1)	3.28 (.1)	-10.46 (-.6)	12.77 (.9)
$n = 5$	-7.89 (-16.8)	-.02 (-.5)	-17.90 (-1.4)	30.25 (1.1)	7.07 (.3)	-11.06 (-.6)	11.83 (.9)

TABLE 4 The χ^2 statistics and p -values in parentheses for various tests of the coefficients of forward rates in specification (S-1). $\neg(\beta_{1,\dots,5}^{(n)} = 0)$ represents negation of $\beta_{1,\dots,5}^{(n)} = 0$ implying one or more of the coefficients $\beta_{1,\dots,5}^{(n)}$ is not zero

Specification: S-1				
$\chi^2 : H_1$	$n = 5$	$n = 4$	$n = 3$	$n = 2$
$\chi^2 : \neg(\beta_{1,\dots,5}^{(n)} = 0)$	26.61 (.00)	35.93 (.00)	22.78 (.00)	31.18 (.00)
$\chi^2 : \beta_5^{(n)} > \beta_1^{(n)}$	5.25 (.02)	4.82 (.03)	2.90 (.09)	.25 (.62)
$\chi^2 : \beta_5^{(n)} > \beta_4^{(n)}$	3.43 (.06)	2.87 (.09)	2.83 (.09)	1.94 (.16)
$\chi^2 : \beta_2^{(n)} > \beta_1^{(n)}$	4.87 (.03)	4.56 (.03)	4.60 (.03)	1.58 (.21)
$\chi^2 : \beta_2^{(n)} > \beta_3^{(n)} > \beta_4^{(n)}$	4.99 (.08)	4.36 (.11)	4.15 (.13)	3.37 (.19)
$\chi^2 : \beta_{1,3,4}^{(n)} = 0, \beta_2^{(n)} > 0, \beta_5^{(n)} > 0$	17.38 (.00)	19.84 (.00)	18.66 (.00)	17.49 (.00)

in conditional volatility. The table also includes tests that compare coefficients $\beta_{1,\dots,5}^{(n)}$ with each other. The tests mostly confirm relative rankings of the coefficients in Figure 1.

Figures 2 and 3 include the plots of the conditional volatilities implied by specifications (S-1) and (S-0) against the magnitudes of the unexpected returns. (S-1) implies high conditional volatilities over the 1980s, a period of high ex post volatility. Overall, (S-1) implies greater variation in the conditional volatility. Specification (S-0) essentially forecasts different average levels of conditional volatilities for different maturities along with a few high-volatility events. The standard deviation of the conditional standard deviation, $\sigma((h_{t+1}^{(n)})^{1/2})$, is one way to quantify the variation in the conditional volatility. The values for the standard deviation of the conditional standard deviation are [.39, .73, .99, and 1.34] for specification (S-1) and [.06, .21, .25, and .33] for specification (S-0) for bonds of maturities $n = 2, 3, 4$, and 5, respectively. The units are percent returns—for example, for the 1-year excess returns (units in percent) of bonds of 2-year maturity, the variation in the conditional standard deviation $(h_{t+1}^{(2)})^{1/2}$ implied by specification (S-1) over the

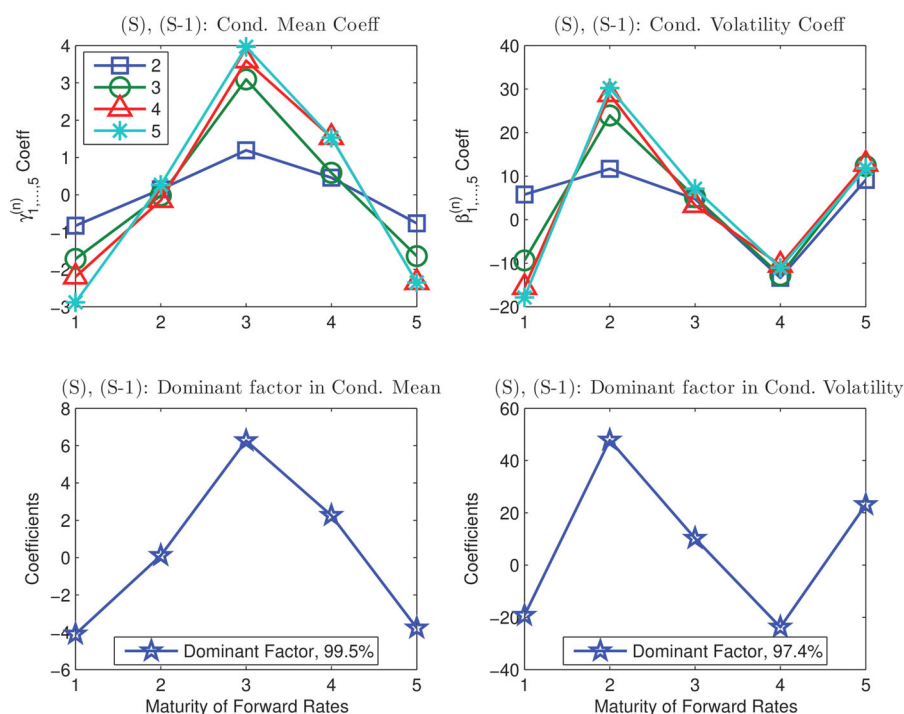


FIGURE 1 Top row: Coefficients of the forward rates in conditional mean (S) and conditional volatility specifications (S-1). Bottom row: Coefficients of the forward rates in the linear combination that defines the dominant factors that capture most of the variation in the conditional means and the conditional volatilities across maturities

entire sample period is of the order of 39 basis points. The conditional volatility implied by specification (S-1) varies much more than that implied by specification (S-0). Thus, the term structure information helps capture a greater amount of variation in the conditional volatility.

Figure 1 plots the coefficient estimates for the specification pair (S), (S-1). A strong tent-shaped pattern is evident in the coefficients of the forward rates in the conditional mean specification (S). In fact, as noted by Cochrane and Piazzesi (2005), a single tent-shaped linear combination of the forward rates captures most of the variation (99.5%) in the conditional means across maturities. I follow Cochrane and Piazzesi (2005) to obtain the coefficients of the forward rates in the linear combination that defines this dominant factor. An eigenvalue decomposition of the covariance matrix of the conditional means yields the loadings of the dominant factor on the conditional means of each of the maturities.

Likewise, a strong single-factor structure is evident in the coefficients of the forward rates in the conditional volatility specification (S-1). A single linear combination of the forward rates captures 97.4% of the variation in the log of conditional volatilities. I use the covariance matrix of the log conditional volatilities to extract the dominant factor.

The patterns of the forward rate coefficients in the linear combination that define the two dominant factors are quite different. In fact, the two factors have a correlation of only .33. Thus, the dominant factor explaining the variation in the conditional mean has little success at explaining the variation in the conditional volatility. The term structure drivers of the conditional mean and the conditional volatility are not the same. I set $\gamma_{1,...,5}^{(n)} = \beta_{1,...,5}^{(n)}$ for each maturity n separately in specification (S-1') forcing the same coefficients for the forward rates in the conditional mean and conditional volatility specifications—that is, the same term structure drivers for the conditional mean and the conditional volatility for the 1-year excess returns of a bond. The drop in the likelihoods or the incremental misfit

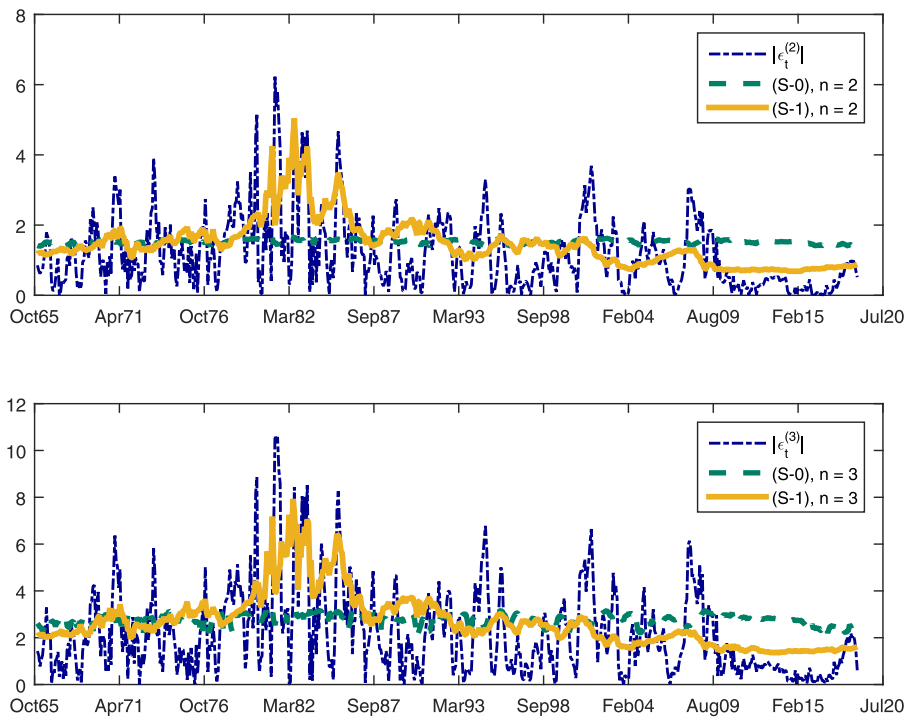


FIGURE 2 Comparison: Magnitude of unexpected returns, $|\epsilon_t^{(\cdot)}|$, conditional volatility implied by specifications (S-0) and (S-1) for the excess returns of bonds of maturities $n = 2$ and 3. All values in percentage, and conditional volatilities in standard deviation

TABLE 5 The χ^2 statistics for the likelihood ratio test of the restricted specification pair (S), (S-1') against the alternate unrestricted specification pair (S), (S-1)

$H_0: S-1' \text{ versus } H_1: S-1$			
	χ^2	p-Value	1% Crit- χ^2
$n = 2$	49.73	.00	15.09
$n = 3$	52.03	.00	15.09
$n = 4$	52.14	.00	15.09
$n = 5$	59.14	.00	15.09

of the specification is one measure of the differences in the term structure drivers of the conditional mean and the conditional volatility.

$$(S): rX_{t+1}^{(n)} = \kappa^{(n)} + \gamma_{1,\dots,5}^{(n)} \cdot f_t^{(1,\dots,5)} + \epsilon_{t+1}^{(n)} \quad \forall n = 2, \dots, 5, \quad (14)$$

$$(S-1'): \ln(h_{t+1}^{(n)}) = \alpha^{(n)} + \delta_1^{(n)} \ln\left(\left(\epsilon_t^{(n)}\right)^2 + \lambda\right) + \gamma_{1,\dots,5}^{(n)} \cdot f_t^{(1,\dots,5)}. \quad (15)$$

The log-likelihoods for the specification pair (S), (S-1') with the restrictions are considerably lower than that for the original specification without the restrictions (S), (S-1). The log-likelihood ratio tests (Table 5) reject the restrictions even at the 1% significance level. Figure 4 compares the conditional means implied by the restricted and unrestricted specification pairs. The expected excess returns implied by the two specifications are almost the same. In fact, the

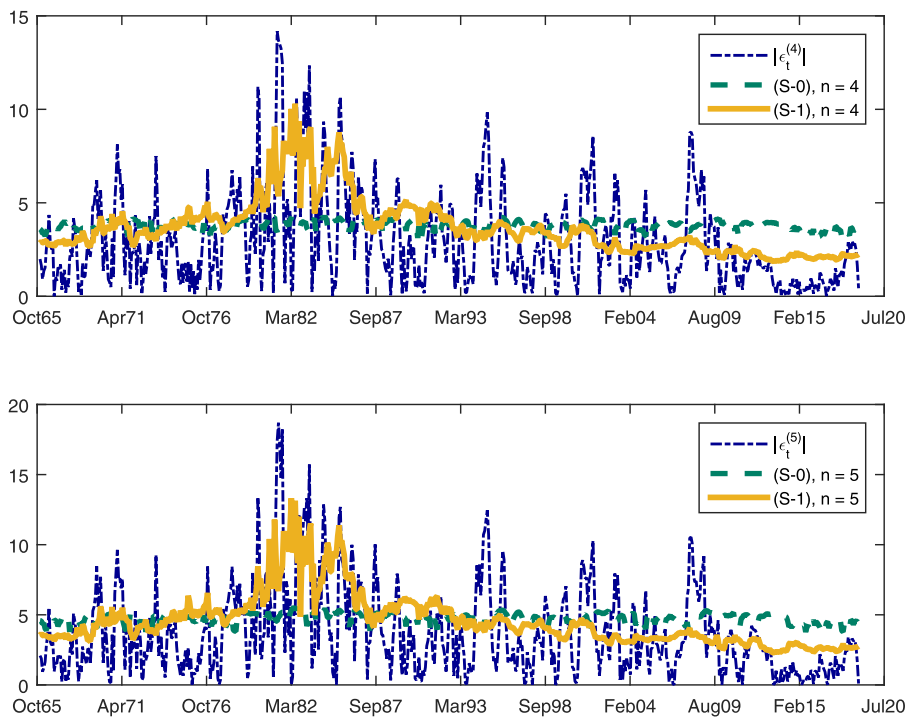


FIGURE 3 Comparison: Magnitude of unexpected returns, $|\epsilon_t^{(n)}|$, conditional volatility implied by specifications (S-0) and (S-1) for the excess returns of bonds of maturities $n = 4$ and 5 . All values in percentage, and conditional volatilities in standard deviation

patterns of the coefficients of the forward rates in the restricted specification pair (S), (S-1') resembles the tent-shaped pattern of the coefficients of the forward rates in the conditional mean part of the unrestricted specification pair (S), (S-1). Figure 5 includes the plots of the conditional volatilities for the restricted and unrestricted specification pairs (for clarity, only the plots for $n = 2$ and 5 are included). The wide fluctuations in the conditional volatilities implied by the unrestricted specification over the sample period are not replicated by the restricted specification. The standard deviations of the conditional standard deviations, $\sigma(h_{t+1}^{(n)})^{1/2}$ are [.06, .22, .25, and .36] for the restricted specification pair and [.39, .73, .99, and 1.34] for the unrestricted specification pair for maturities $n = [2, 3, 4, \text{ and } 5]$, respectively. The unrestricted specification pair captures much more variation in the conditional volatility than the restricted specification pair, whereas the conditional means implied by the two specifications are almost the same. Thus, forcing the same term structure drivers for the conditional mean and the conditional volatility limits the ability to forecast the conditional volatility. The maximum likelihoods are substantially lower for the restricted specification pair compared to the unrestricted specification pair. Thus, the term structure drivers of the conditional mean and the conditional volatility are different.

Thus, the term structure captures a significant amount of variation in the conditional volatility of excess returns. A single linear combination of the forward rates captures most of the variation in the log conditional volatilities due to the variation in the forward rates. Hence, the information in the term structure for the variation in the conditional volatility is summarized by a single linear combination of forward rates. However, the term structure drivers of the conditional mean and the conditional volatility are different. A restriction to force the same term structure drivers is rejected in the data. Thus, two different linear combinations of the forward rates summarize the variation in the Sharpe ratios of 1-year excess returns of bonds of 2- to 5-year maturities. The tent-shaped Cochrane–Piazzesi factor that captures the expected excess returns captures the aggregate price of risk and not the conditional volatility of 1-year bond returns.

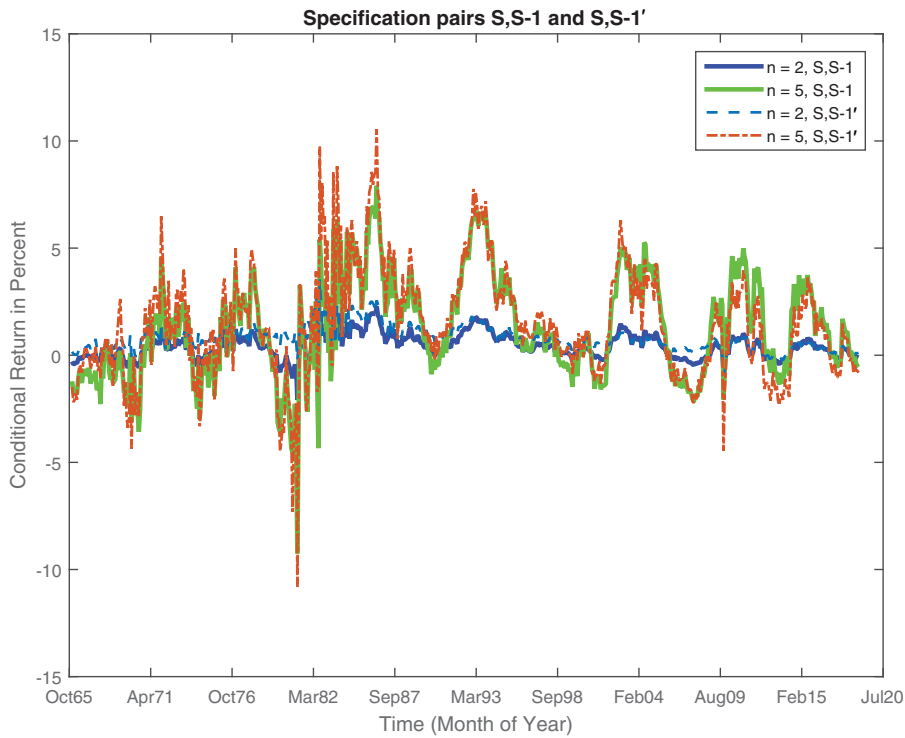


FIGURE 4 Comparison of the conditional means implied by specification pairs (S), (S-1) and (S), (S-1'). For clarity only, the plots for $n = 2$ and 5 are plotted. The two plots for each n are on top of each other and hard to distinguish. The coefficients of the forward rates in the conditional volatility specification (S-1') are restricted to be the same as the coefficients of the forward rates in the conditional mean specification (S). Returns are given in percentage

TABLE 6 The joint significance test for the coefficients of forward rates in the conditional volatility specification (S-1) (conditional mean specification (S))

Specification: S-1, $H_0: \sum_{i=1}^5 \beta_i^{(n)} = 0$			
	χ^2	p-Value	1% Crit- χ^2
$n = 2$	20.44	.00	6.63
$n = 3$	25.30	.00	6.63
$n = 4$	25.04	.00	6.63
$n = 5$	31.45	.00	6.63

5 | INFORMATION IN THE LEVEL

In Section 4, Figures 2 and 3 plot the conditional volatilities implied by specification (S-1). Note that the high conditional volatilities over the 1980s match with the high ex post volatility over the same period. This match could very well be just due to the level effect. And the information in the term structure for conditional volatility estimation could just be due to the level of interest rates. In this section, I find that, in fact, most of the variation in the conditional volatilities is captured by the overall level of interest rates.

One way to check whether the level of interest rates has any contribution toward conditional volatility estimation is to check whether the sum of the coefficients of forward rates in (S-1) is zero. The χ^2 statistics in Table 6 reject the

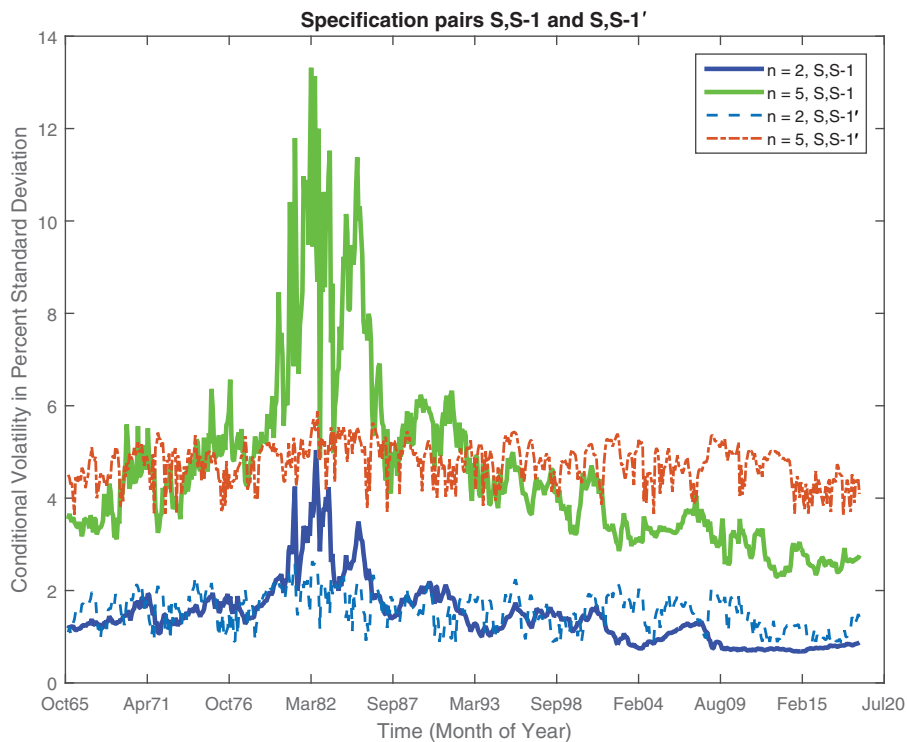


FIGURE 5 Comparison of the conditional volatilities implied by specification pairs (S), (S-1) and (S), (S-1'). For clarity only, the plots for $n = 2$ and 5 are included. The coefficients of the forward rates in the conditional volatility specification (S-1') are restricted to be the same as the coefficients of the forward rates in the conditional mean specification (S). Volatility is in percentage standard deviation

hypothesis that the sum of the forward rate coefficients is zero. The null is rejected even at the 1% significance level. Thus, a rise in the overall level of interest rates increases the conditional volatility of 1-year excess returns of bonds.

$$(S-2): h_{t+1}^{(n)} = \exp \left(\alpha^{(n)} + \delta_1^{(n)} \ln \left(\left(\epsilon_t^{(n)} \right)^2 + \lambda \right) + \beta_1^{(n)} f_t^{(1)} + \beta_{2,\dots,5}^{(n)} \cdot \left(f_t^{(2,\dots,5)} - f_t^{(1)} \right) \right), \quad (16)$$

$$(S-3): h_{t+1}^{(n)} = \exp \left(\alpha^{(n)} + \delta_1^{(n)} \ln \left(\left(\epsilon_t^{(n)} \right)^2 + \lambda \right) + \beta_{2,\dots,5}^{(n)} \cdot \left(f_t^{(2,\dots,5)} - f_t^{(1)} \right) \right), \quad (17)$$

$$(S-4): h_{t+1}^{(n)} = \exp \left(\alpha^{(n)} + \delta_1^{(n)} \ln \left(\left(\epsilon_t^{(n)} \right)^2 + \lambda \right) + \beta_1^{(n)} f_t^{(1)} \right). \quad (18)$$

I test the hypothesis that the information in the level of interest rates captures a significant amount of variation in the conditional volatility of excess returns. I use specification (S-2), which separates the overall level from the spreads in a parsimonious way. The short-term rate proxies for the overall level of interest rates in (S-2).

The coefficient estimates and the standard errors for the level and spread specification (S-2) are in the Appendix in Supporting Information. Representative extracts for the bonds of 4-year maturity ($n = 4$) are in Tables 7 and 8. The coefficients for the short-term rate are positive and overwhelmingly significant. Thus, a higher overall level of interest rates bodes higher conditional volatility for the bond excess returns. The significance of the coefficients of the short rate imply that the information in the level of interest rates captures a considerable amount of the variation in conditional volatilities. In fact, going by the individual t -ratios, level seems to be the dominant contribution of the term structure of interest rates for predicting the conditional volatilities.

TABLE 7 Specification pair: (S), (S-2). Coefficients and the t-ratios for the conditional mean part of the specification (S) for the 1-year excess return of 4-year maturity bond ($n = 4$)

$rx_{t+1}^{(4)} = \kappa^{(4)} + \gamma_{1,...,5}^{(4)} \cdot f_t^{(1,...,5)} + \epsilon_{t+1}^{(4)}$						
	$\kappa^{(4)}$	$\gamma_1^{(4)}$	$\gamma_2^{(4)}$	$\gamma_3^{(4)}$	$\gamma_4^{(4)}$	$\gamma_5^{(4)}$
Coeff	−0.04	−2.20	−.15	3.59	1.54	−2.35
t-Ratio	−5.26	−6.90	−.24	6.09	3.15	−6.45

TABLE 8 Specification pair: (S), (S-2). Coefficients and the t-ratios for the conditional volatility part of the specification (S-2) for the 1-year excess return of 4-year maturity bond ($n = 4$)

$\ln \left(h_{t+1}^{(4)} \right) = \alpha^{(4)} + \delta_1^{(4)} \ln \left(\left(\epsilon_t^{(4)} \right)^2 + \lambda \right) + \beta_1^{(4)} f_t^{(1)} + \beta_{2,\dots,5}^{(4)} \cdot \left(f_t^{(2,\dots,5)} - f_t^{(1)} \right)$
$\alpha^{(4)} \qquad \delta_1^{(4)} \qquad \beta_1^{(4)} \qquad \beta_2^{(4)} \qquad \beta_3^{(4)} \qquad \beta_4^{(4)} \qquad \beta_5^{(4)}$
Coeff −8.19 −.02 18.82 28.77 3.27 −10.45 12.77
t-Ratio −16.69 −.52 5.00 1.10 .14 −.57 .93

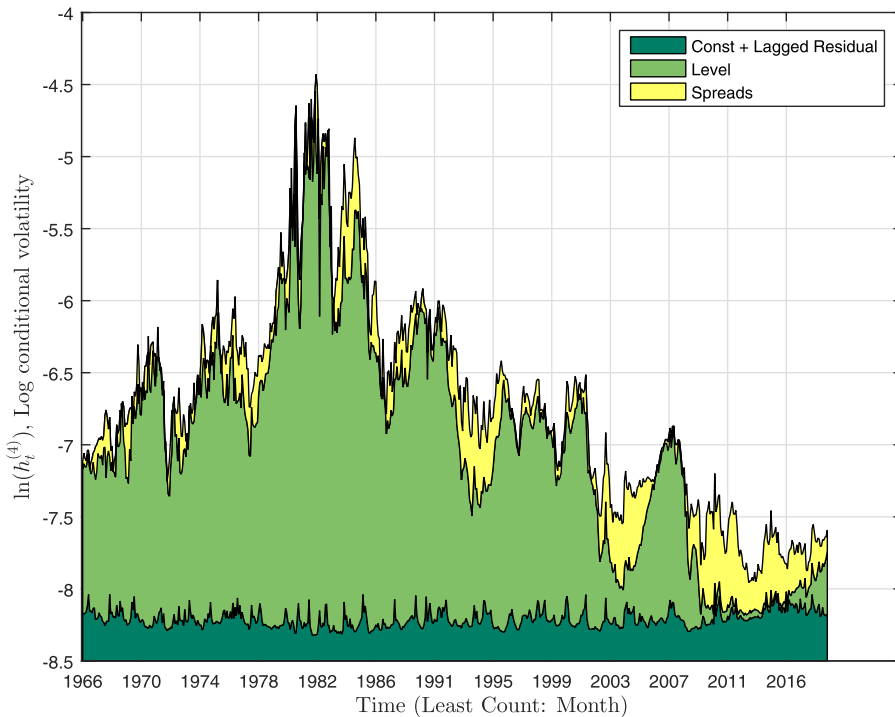


FIGURE 6 Decomposition of log conditional volatility for $n = 4$ (specification (S-2)) into three components: constant and lagged residual: $\alpha^{(4)} + \delta_1^{(4)} \times \ln(\epsilon_t^{(4)2} + \lambda)$; level: $\beta_1^{(4)} \times f_t^{(1)}$; and spreads: $\beta_{2,\dots,5}^{(4)} \cdot (f_t^{(2,\dots,5)} - f_t^{(1)})$. The baseline is shifted to -8.5 instead of zero for convenience

Figure 6 depicts the contribution of the short-term rate toward predicting the conditional volatility. The level-and-spread specification (S-2) for the log of conditional volatility is linear in the short rate, the spreads and the log of lagged residual plus a constant. Figure 6 shows that the variation in the short-term rate is a significant component of the variation of log conditional volatility. In fact, the short rate dominates the variation, and the spreads capture the second most significant component of the variation.

TABLE 9 The maximum log-likelihood values for different specifications. The term structure variables in each conditional volatility specification: (S-1): all forward rates, (S-2): short rate and all the spreads, (S-3): all the spreads, and (S-4): short rate

Log-likelihood values for different specifications				
	S-1	S-2	S-3	S-4
$n = 2$	1,335.44	1,335.44	1,319.51	1,333.97
$n = 3$	1,055.74	1,055.74	1,037.22	1,049.85
$n = 4$	911.86	911.86	892.92	904.15
$n = 5$	809.69	809.69	787.39	799.98

TABLE 10 The standard deviation of the conditional standard deviation implied by different specifications. The term structure drivers of the conditional volatility specification are (S-1): all forward rates, (S-2): short rate and all the spreads, (S-3): all the spreads, and (S-4): short rate. Units: percentage returns

$\sigma \left((h_{t+1}^{(n)})^{1/2} \right)$ for different specifications				
	$n = 2$	$n = 3$	$n = 4$	$n = 5$
S-1	.39	.73	.99	1.34
S-2	.39	.73	.99	1.34
S-3	.24	.40	.53	.78
S-4	.39	.67	.86	1.14

Another way to check the importance of the short rate is to quantify the impact of the zero coefficient restriction on the short rate for the maximized value of the log-likelihood. Specification (S-3) employs only the spreads for the conditional volatility estimation and sets the coefficient on the short rate to zero. Table 9 compares the maximized log-likelihood values for different specifications. A move from the simplified conditional volatility specification (S-1) to the level-and-spread specification (S-2) does not alter the log-likelihoods. However, dropping the short rate and keeping only the spreads (specification (S-3)) significantly reduces the maximized log-likelihood values. The likelihood ratio tests reject the zero coefficient restriction on the short rates even at the 1% significance level. Thus, the overall level of interest rates captures a substantial component of the variation of the conditional volatilities of 1-year excess returns.

Table 10 presents $\sigma((h_{t+1}^{(n)})^{1/2})$, the standard deviations of the conditional standard deviations implied by different specifications. No noticeable differences exist between the $\sigma((h_{t+1}^{(n)})^{1/2})$ values implied by specification (S-1) and those implied by specification (S-2). Thus, a move to level-and-spread representation of the term structure information does not alter the variation captured by the conditional volatility specification. However, the drop in the standard deviation of the conditional standard deviation is substantial once the short rate is removed from the specification ((S-2) vs. (S-3)). The drop in $\sigma((h_{t+1}^{(n)})^{1/2})$ is another measure of the importance of the overall level for predicting the conditional volatility. Also of interest is the drop in the variation once all the spreads are removed. The drop in the variation due to the absence of the spreads ((S-2) vs. (S-4)) is minimal compared to the drop in the absence of the short rate ((S-2) vs. (S-3)). In fact, the absence of the spreads hardly affects the variation in the conditional volatility of the 1-year excess returns of bonds of 2-year maturity. Thus, the short rate captures a significantly greater amount of variation of the conditional volatility than the spreads.

The nominal cash flows of bonds are fixed, and as a result, the news about inflation affects the excess bond returns. Fischer (1981) shows that the volatility in inflation increases with the level. To the extent that the short rate captures the expectations about inflation, a higher short rate predicts higher conditional volatility for excess bond returns as implied by the conditional volatility specification (S-2).

TABLE 11 Likelihood ratio test of zero coefficient restriction on the spreads. (S-4) sets coefficients of spreads to zero, whereas (S-2) employs both the level and the spreads to predict the conditional volatility

H_0 : S-4 versus H_1 : S-2			
	χ^2	p-Value	1% Crit- χ^2
$n = 2$	2.95	.57	13.28
$n = 3$	11.77	.02	13.28
$n = 4$	15.42	.00	13.28
$n = 5$	19.41	.00	13.28

6 | INFORMATION IN SPREADS

In this section, I check whether the spreads capture any significant amount of variation in the conditional volatility. I find that a model without the spreads is inferior relative to a model with the spreads. In other words, spreads do carry a nontrivial amount of information for forecasting the conditional volatility. I reduce the number of spreads in the model to just two spreads. Further, I find that the 4-5 yield spread has opposing effects on the conditional mean and the conditional volatility.

In Section 5, Figure 6 shows that the overall level of interest rates captures a significant amount of the variation in the conditional volatility of excess returns. I compare the log-likelihoods for specification (S-4), which contains only the short rate representing the information in the term structure against the log-likelihoods for specification (S-2), which contains both the short rate and the spreads. Specification (S-4) sets zero coefficient restriction on the spreads. Table 11 presents the likelihood ratio statistic for the zero coefficient restriction on the spreads. The drop in the maximum log-likelihoods is significant, except for the bonds of 2-year maturity. However, the coefficients of the spreads are not jointly significant (level of significance is between 12% for $n = 5$ to 36% for $n = 2$). The implications of the χ^2 Wald test and the likelihood ratio test differ. This is because the χ^2 Wald test and the likelihood ratio tests are only asymptotically equivalent.

The standard deviation of the conditional standard deviation $\sigma((h_{t+1}^{(n)})^{1/2})$ is, however, lower once the spreads are dropped from the specification, except for $n = 2$. As noted in Table 10, values for $\sigma((h_{t+1}^{(n)})^{1/2})$ for specification (S-4) are [.39, .67, .86, and 1.14] compared to the values [.39, .73, .99, and 1.34] for specification (S-2). Thus, the absence of spreads does drop the captured amount of variation in the conditional volatility. Thus, the spreads do carry a nontrivial amount of information useful for predicting the conditional volatility.

I drop the spreads that do not significantly contribute toward predicting the conditional volatility. The tiny t -ratios for the coefficients of the spreads in the conditional volatility specification (S-2) are suggestive of near-multicollinearity. The middle row of Figure 7 plots the coefficients of the forward rates, short rate, and the spreads for the specification pair (S), (S-2). The sharp changes in the signs of the coefficients of the spreads in the conditional volatility specification suggest near-multicollinearity.

I use the log-likelihood values, the standard deviation of conditional standard deviation, and Wald tests to guide through the specification search and choose the appropriate subset of spreads that contribute the most toward predicting the conditional volatility. I refer to the spreads between the forward rate with maturity n and the short rate ($n = 1$) as the $n - 1$ spread. Dropping the 5-1 spread results in a significant drop in the log-likelihood values, whereas dropping the 3-1 spread results in no noticeable differences. The final reduced specification is (S-6) with the short rate and the 4-1 and 5-1 spreads. Details of the specification search are in the Appendix in Supporting Information:

$$(S-2): h_{t+1}^{(n)} = \exp \left(\alpha^{(n)} + \delta_1^{(n)} \ln \left(\left(\epsilon_t^{(n)} \right)^2 + \lambda \right) + \beta_1^{(n)} f_t^{(1)} + \beta_{2,\dots,5}^{(n)} \cdot \left(f_t^{(2,\dots,5)} - f_t^{(1)} \right) \right), \quad (19)$$

$$(S-6): h_{t+1}^{(n)} = \exp \left(\alpha^{(n)} + \delta_1^{(n)} \ln \left(\left(\epsilon_t^{(n)} \right)^2 + \lambda \right) + \beta_1^{(n)} f_t^{(1)} + \beta_{4,5}^{(n)} \cdot \left(f_t^{(4,5)} - f_t^{(1)} \right) \right). \quad (20)$$

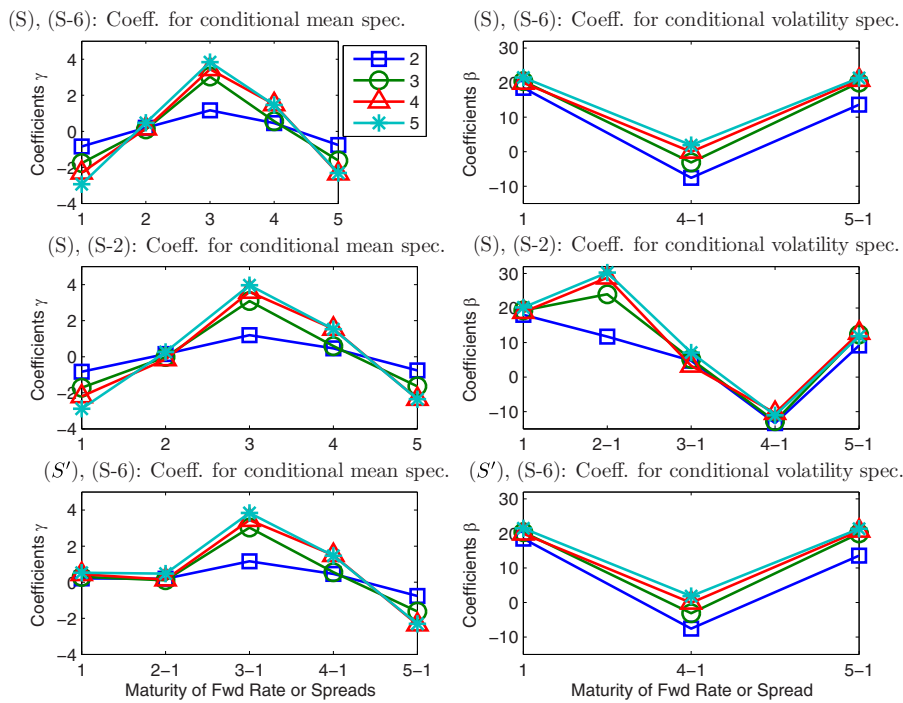


FIGURE 7 Conditional mean specification (S) (top left) and conditional volatility specification (S-6) (top right). Conditional mean specification (S) (middle left) and conditional volatility specification (S-2) (middle right). Conditional mean specification (S') (bottom left) and conditional volatility specification (S-6) (bottom right)

TABLE 12 The specification pair (S), (S-6). The coefficient estimates and the *t*-ratios for the conditional volatility part of specification (S-6)

	$\ln(h_{t+1}^{(n)}) = \alpha^{(n)} + \delta_1^{(n)} \ln\left(\left(\epsilon_t^{(n)}\right)^2 + \lambda\right) + \beta_1^{(n)} f_t^{(1)} + \beta_4^{(n)} \cdot \left(f_t^{(4,5)} - f_t^{(1)}\right)$				
	$\alpha^{(n)}$	$\delta_1^{(n)}$	$\beta_1^{(n)}$	$\beta_4^{(n)}$	$\beta_5^{(n)}$
$n = 2$	-10.05 (-19.2)	-0.04 (-1.0)	18.50 (5.0)	-7.57 (-6)	13.60 (1.3)
$n = 3$	-9.23 (-20.9)	-0.06 (-1.9)	20.40 (5.5)	-3.10 (-2)	20.01 (1.8)
$n = 4$	-8.21 (-17.1)	-0.01 (-0.3)	19.92 (5.6)	-0.06 (-0)	20.75 (1.9)
$n = 5$	-7.95 (-17.8)	-0.02 (-0.4)	21.48 (6.3)	1.87 (.1)	21.36 (1.9)

The top row of Figure 7 plots the coefficients for the specification pair (S), (S-6). The coefficient estimates and the *t*-ratios for the conditional volatility part of the specification pair are in Table 12. The short rate is highly significant for all the maturities, and the *t*-ratios of the 5-1 spreads are between a low of 1.28 and a high of 1.98. The coefficients of the 4-1 spread are not significant. However, the coefficients of the 4-1 and 5-1 spreads are jointly significant except for $n = 2$ (significance levels for $n = 2, 3, 4$, and 5 are [34%, 8%, 2%, and 0%], respectively). The coefficients for the conditional mean part of the specification pair (S), (S-6) are in Table 23 in the Appendix in Supporting Information:

$$(S') : r x_{t+1}^{(n)} = \kappa^{(n)} + \gamma_1^{(n)} f_t^{(1)} + \gamma_{2,\dots,5}^{(n)} \cdot \left(f_t^{(2,\dots,5)} - f_t^{(1)}\right) + \epsilon_{t+1}^{(n)} \quad \forall n = 2, \dots, 5. \quad (21)$$

A higher level of overall interest rates implies higher conditional volatility. To extract the effect of the level of interest rates on the conditional mean, I use (S') for the conditional mean specification along with ($S-6$) for the conditional volatility specification. Specification (S') captures the term structure information in the level and spreads. The short rate proxies for the level of interest rates in (S'). The bottom row of plots in Figure 7 corresponds to this combination. The coefficient on the short rate is positive and significant for all maturities in (S'). The t -ratios (not included) indicate that the spreads also capture significant amount of the variation in the conditional mean and the short rate does not overwhelm the significance of spreads. Thus, the conditional volatility and the conditional mean go in the same direction with respect to the overall level of interest rates.

Cochrane and Piazzesi (2005) find that a single tent-shaped factor of forward rates captures the risk premium (or the expected excess returns) associated with the 1-year returns on bonds of 2- to 5-year maturity. A similar tent-shaped factor is evident in the coefficients of the forward rates in the conditional mean specification (S) (Figure 7). They also note that the tiny 4-5 yield spread is important for forecasting all maturity bond returns. The high positive coefficients on the 5-1 spread in Figure 7 indicate that the 5-4 year yield spread (or the negative of the 4-5 yield spread) forecasts higher conditional volatility for the 1-year excess returns of all maturities. Thus, the 4-5 yield spread has opposing effects on the conditional volatility and conditional mean.

However, the opposing effect of the 4-5 yield spread does not translate into a pronounced negative correlation between the conditional volatility and the conditional mean. This is because the variation in the conditional volatility is dominated by the variation in the level or the short rate and not the spreads. Also, the coefficients on the 2-1 and 3-1 spreads are positive in the conditional mean specification. The strong positive correlation between the 5-1 spread and the 2-1 and 3-1 spreads (.85 and .90, respectively) subdues a strong potential negative correlation between the drivers of the conditional mean and the conditional volatility.

The opposing effect of the 4-5 yield spread on the conditional mean and the conditional volatility is in line with the conclusion in Section 4 that the term structure drivers of the conditional mean and the conditional volatility are different. Most of the variation in the conditional volatility is due to variation in the overall level of interest rates. However, the short rate does not overwhelm the spreads' contribution to the variation in the conditional mean. Thus, the drivers of the conditional mean and the conditional volatility are different. In fact, forcing the same coefficients for the short rate and the 4-1 and 5-1 spreads in the conditional mean and the conditional volatility specifications for each maturity, n , results in a severe misfit to the data. Similar to the results in Section 4, the likelihood ratio tests resoundingly reject this restriction.² Thus, the information in the term structure for the variation in the Sharpe ratios of 1-year returns of bonds of 2- to 5-year maturities is summarized by two different drivers. The variation in the expected excess returns captures variation in the market price of risk and not the variation in the conditional volatility of 1-year bond returns.

7 | EFFECTS OF CHANGE IN MACROECONOMIC VOLATILITY

The interest rate data stretch over 1964–2018. The U.S. economy experienced a generally increasing short-term interest rate and increasing inflation until mid 1981 and an equally declining short-term interest rate and inflation over the rest of the sample period. Glosten et al. (1993) suggest that the short rate also reflects aggregate economic uncertainty in addition to the future expected inflation. Kim and Nelson (1999) and McConnell and Perez-Quiros (2000) show that the post-1981 period is characterized by significantly lower real and nominal macroeconomic volatility. I verify whether these differences between the pre- and post-1980s alter any of the conclusions about term structure drivers of the conditional volatility.

I pick the pre-1980s sample spanning the period January 1964 to December 1979, and conduct various tests similar to those in earlier sections and verify if the conclusions change. First, I estimate the specification pair (S),

²Although the variation in the conditional volatility implied by the restricted specification pair drops, the conditional means implied by the restricted and unrestricted specification pairs are the same. Results are similar to those obtained for the restricted specification pair (S), ($S-1'$).

TABLE 13 The first two rows present χ^2 statistics and p -values in parentheses for various tests of the coefficients of forward rates in specification (S-1). $\neg(\beta_{1,\dots,5}^{(n)} = 0)$ represents negation of $\beta_{1,\dots,5}^{(n)} = 0$ implying one or more of the coefficients $\beta_{1,\dots,5}^{(n)}$ is not zero. The bottom two rows present χ^2 statistics and p -values in parentheses for likelihood ratio tests comparing conditional volatility specifications (S-0) versus (S-1) and (S-1') versus (S-1). The same conditional mean specification (S) applies in all these cases. The estimates are based on data spanning until December 1979

Conditional volatility specifications: S-1, S-0, and S-1'				
$\chi^2 : H_1$	$n = 5$	$n = 4$	$n = 3$	$n = 2$
χ^2 (S-1): $\neg(\beta_{1,\dots,5}^{(n)} = 0)$	23.23 (.00)	33.93 (.00)	22.78 (.00)	35.97 (.00)
χ^2 (S-1): $\sum_{i=1}^5 \beta_i^{(n)} \neq 0$	18.82 (.00)	16.56 (.00)	17.95 (.00)	17.71 (.00)
χ^2 (S-0 vs. S-1): $H_1: S-1$	33.92 (.00)	29.09 (.00)	41.68 (.00)	39.31 (.00)
χ^2 (S-1' vs. S-1): $H_1: S-1$	25.82 (.00)	27.99 (.00)	38.83 (.00)	38.74 (.00)

(S-1) with this limited data. I present tests on coefficient estimates from specification (S-1) in Table 13. The values and t -ratios of coefficient estimates are in Table 24 in the Appendix in Supporting Information. Similar to the full-sample case, the coefficients of forward rates in (S-1) are not individually significant but jointly significant (first row in Table 13).

The specification (S-0), which lacks any term structure information, is rejected in favor of (S-1) in the likelihood ratio test (third row in Table 13). Thus, similar to the full-sample case, the term structure captures a nontrivial component of the variation in conditional volatility. Further, the likelihood ratio test in the last row in Table 13 rejects specification (S-1') in favor of (S-1). (S-1') sets the same coefficients for the forward rates in conditional volatility specification as that in conditional mean ($\gamma_{1,\dots,5}^{(n)} = \beta_{1,\dots,5}^{(n)}$) specification (S). This test, similar to the full-sample case, reveals that the term structure drivers of conditional mean and conditional volatility are distinct and not the same. Therefore, the Cochrane–Piazzesi factor captures aggregate price of risk and not the amount of bond return risk even in the data sample limited to December 1979.

The test of $\sum_{i=1}^5 \beta_i^{(n)} = 0$ for $\beta_i^{(n)}$ in (S-1) is summarily rejected (second row in Table 13). Hence, same as in the full-sample case, level of interest rates is essential for capturing the variation in conditional volatility. I also estimate specification pair (S), (S-2) with the limited data sample. Specification (S-2), which separates level and spread term structure drivers, helps assess relative importance of these for conditional volatility. The coefficient estimates and t -ratios for (S-2) are in Table 24 in the Appendix in Supporting Information. Additionally, the comparisons of specification (S-2) with (S-3) and (S-4) are included in Table 25 in the Appendix in Supporting Information. Similar to the full sample, these tests reveal that interest rate level captures most of the variation in conditional volatility, whereas importance of spreads for explaining conditional volatility earns mixed evidence. The Appendix in Supporting Information includes more tests and analysis of reduced specification (S-6) as well. In summary, the tests reveal that the conclusions in earlier sections do not change in the pre-1980s sample.

The Appendix in Supporting Information also includes forecasting exercise to assess the impact of forward-looking bias on full-sample conditional volatility estimates. I conduct this analysis using both the expanding and rolling window methods. Both methods are able to capture the high level of volatilities during the 1980s and show a close association with full-sample estimates of conditional volatilities. These results show that the impact of forward-looking bias on full-sample estimates is minimal and do not alter the conclusions based on full-sample analysis.

8 | LAGGED CONDITIONAL VOLATILITY AND ARCH/GARCH MODELS

In this section, I consider the conditional volatility specification that includes the lagged conditional volatility as one of the explanatory variables. I also consider the ARCH/GARCH models of conditional volatility. I find that the lagged conditional volatility does not significantly add to the ability to predict the future conditional volatility:

$$(S-1): h_{t+1}^{(n)} = \exp \left(\alpha^{(n)} + \delta_1^{(n)} \ln \left(\left(\epsilon_t^{(n)} \right)^2 + \lambda \right) + \beta_{1,\dots,5}^{(n)} \cdot f_t^{(1,\dots,5)} \right), \quad (22)$$

$$(\overline{S-1}): h_{t+1}^{(n)} = \exp \left(\alpha^{(n)} + \delta_1^{(n)} \ln \left(\left(\epsilon_t^{(n)} \right)^2 + \lambda \right) + \eta_1^{(n)} \ln \left(h_t^{(n)} \right) + \beta_{1,\dots,5}^{(n)} \cdot f_t^{(1,\dots,5)} \right), \quad (23)$$

$$(S-0): h_{t+1}^{(n)} = \exp \left(\alpha^{(n)} + \delta_1^{(n)} \ln \left(\left(\epsilon_t^{(n)} \right)^2 + \lambda \right) \right), \quad (24)$$

$$(\overline{S-0}): h_{t+1}^{(n)} = \exp \left(\alpha^{(n)} + \delta_1^{(n)} \ln \left(\left(\epsilon_t^{(n)} \right)^2 + \lambda \right) + \eta_1^{(n)} \ln \left(h_t^{(n)} \right) \right). \quad (25)$$

The conditional volatility specification $(\overline{S-1})$ nests specification $(S-1)$ and also uses the lagged conditional volatility to forecast the conditional volatility in the future. Likewise, $(\overline{S-0})$ adds lagged conditional volatility to the conditional volatility specification $(S-0)$. A comparison of $(\overline{S-0})$ and $(\overline{S-1})$ is similar to the comparison of $(S-0)$ and $(S-1)$ in Section 4, overwhelmingly rejecting the zero coefficient restrictions on the forward rates. The term structure information does improve the ability to forecast the conditional volatility of 1-year excess returns. Figure 9 in the Appendix in Supporting Information plots the conditional volatilities implied by specifications $(\overline{S-1})$ and $(\overline{S-0})$ against the absolute value of unexpected excess returns $\epsilon_{t+1}^{(n)}$ for $n = 2$ and 5 .

The lagged conditional volatility, however, does not significantly add to the ability to predict the conditional volatility in the future. The coefficients of the lagged conditional volatilities $\eta_1^{(n)}$ in specification $(\overline{S-1})$ are not significant (t -ratios vary between .01 and .67). Also, the likelihood ratio tests fail to reject a zero coefficient restriction on the lagged conditional volatility.

The standard deviation of the conditional standard deviation of the excess returns measures the variation in the conditional volatility implied by the model. The values of the standard deviation of the conditional standard deviation $\sigma((h_{t+1}^{(n)})^{1/2})$ are $[39.1, 72.9, 99.4, \text{ and } 133.7] \times 10^{-2}$ for $(S-1)$ and $[39.4, 72.9, 99.9, \text{ and } 133.7] \times 10^{-2}$ for $(\overline{S-1})$ for bonds of maturities $n = [2, 3, 4, \text{ and } 5]$, respectively. The units are percent returns—for example, for the 1-year excess returns (units in percent) of bonds of 2-year maturity, the variation in the conditional standard deviation $(h_{t+1}^{(2)})^{1/2}$ over the entire sample period is of the order of 39 basis points. The two sets of values are close and differ only in the third decimal (1/10 of a basis point). Thus, as suggested by the likelihood ratio tests, including the lagged conditional volatility does not significantly improve the ability to capture the variation in the conditional volatility. More tests with standard ARCH/GARCH models but yielding the same conclusion are in the Appendix in Supporting Information. Hence, I use the specifications without the lagged conditional volatility throughout the paper.

9 | CONCLUSION

I find time variation in the conditional volatility of 1-year excess returns of bonds. Information in the term structure captures a significant amount of the variation in the conditional volatilities of 1-year excess returns. A single linear combination of the forward rates captures most of the variation in the conditional volatilities of 1-year excess returns of bonds of 2- to 5-year maturities. The term structure drivers of the conditional mean and the conditional volatilities are not the same. Two different linear combinations of the forward rates summarize the information in the term structure for the Sharpe ratios of 1-year excess returns of bonds of 2- to 5-year maturities. An attempt to force the same

term structure drivers for the conditional mean and the conditional volatility is rejected in the data. Thus, the tent-shaped Cochrane–Piazzesi factor that captures the expected excess returns proxies the market price of risk but not the conditional volatility of 1-year bond return. Therefore, it can be used to explain cross-section of asset returns.

The change in the overall level of interest rates is the most important component of term structure affecting the conditional volatility. Spreads of the forward rates relative to the short rate capture some additional variation in the conditional volatility. Although spreads do not capture a large portion of the variation in the conditional volatility, their contribution is nontrivial. The variation in spreads is mostly due to the 4–5 yield spread that has a negative relation with the conditional volatility but a positive relation with the conditional mean.

The contribution of the short rate toward conditional volatility is best interpreted in light of increasing inflation uncertainty with the level of inflation (Fischer, 1981). To the extent that the short rate captures future inflation expectations, a higher short rate forecasts increasing uncertainty or the conditional volatility for the excess returns of bonds.

Thus, I find that the term structure drivers of the conditional mean and the conditional volatility of 1-year excess bond returns are different. One linear combination of the forward rates for the conditional mean and a different one for the conditional volatility summarize the variation in these two moments. The variation in the tent-shaped Cochrane–Piazzesi factor (Cochrane and Piazzesi, 2005) captures the market price of risk but not the conditional volatility of 1-year bond returns. This conclusion and the evidence in this paper indicate that the Cochrane–Piazzesi factor can be used for cross-sectional asset pricing.

ACKNOWLEDGMENTS

I thank Prof. John Cochrane and Prof. Lars Hansen for numerous discussions and comments. I also thank Professors Chinmay Jain, Monika Piazzesi, Jeffrey R. Russell, Ruey S. Tsay, Pietro Veronesi, participants at Financial Management Association International 2014 meetings, and seminar participants at the University of Chicago Booth School of Business for valuable comments. I thank Alexi Savov for numerous proof reads.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

How to cite this article: Khanapure RB. Term structure determinants of time-varying risk of 1-year bond returns. *Financial Review*. 2020;55:365–384. <https://doi.org/10.1111/fire.12222>